

Kim-Anh Le

📍 Hanoi, Vietnam ✉ anhlkchloe@gmail.com ☎ (+84)-914738168 🆔 0009-0000-4038-3753 in LinkedIn

Education

National Economics University

Bachelor of Mathematical Economics - High Distinction

Hanoi, Vietnam

Oct 2021 – Feb 2025

- **Overall GPA:** 3.65/4.0 (8.71/10.0)
- **Graduation Thesis:** *Analysis of Optimal Tax Implementation on Polluting Fuels in General Equilibrium*, (Best Thesis: 9.8/10.0).

Lam Son High School for the Gifted

Mathematics Specialization

Thanh Hoa, Vietnam

Jul 2018 – May 2021

Research Experience

Quantitative Resources for Economic and Financial Modeling Lab

Research Assistant

Hanoi, Vietnam

Aug 2023 – Jun 2025

Direct Supervisor: Dr. Van-Quy Nguyen

Project: Overlapping Generation Model with Wariness

- Formulated and resolved a two-period household utility maximization problem featuring a non-smooth max-min objective function capturing lifecycle wariness and derived a non-linear, piecewise capital law of motion by analyzing binding conditions across three endogenous return-driven regimes.
- Constructed a global phase diagram for a logarithmic Cobb-Douglas economy using the qualitative theory of dynamical systems, mapping individual behavioral regimes to aggregate capital paths to prove steady-state existence, uniqueness, and finite-threshold projection.
- Analyzed dynamic efficiency and derived precise parametric boundaries for decentralized golden-rule implementation, further extended the framework to a Romer-style endogenous growth model to mathematically demonstrate asymptotic growth saturation and finite-threshold stabilization.

Project: Existence and Regularity of Competitive Equilibria in Economies with Externalities

- Constructed a comprehensive general equilibrium framework for a pure exchange economy integrating non-separable consumption and wealth externalities across multi-commodity spaces (L goods, m consumers)
- Formulated a C^3 differential topology framework and successfully proved the global existence of competitive equilibria using a multi-dimensional continuous deformation homotopy system ($H(\xi, t) = 0$).
- Applied the Transversality and Regular Value Theorems to establish generic regularity, proving that regular economies form an open and full Lebesgue measure subset, where the equilibrium set $F_e^{-1}(0)$ is finite (odd) and local equilibria depend differentiably (C^1) on economic parameters.

Research Student

Personal Project: Investment Strategies with Habit Formation under Mispricing

Hanoi, Vietnam

Oct 2025 – Present

- Derived exact closed-form feedback control strategies for optimal consumption and investment under CRRA and logarithmic utilities, mathematically decomposing the asset allocation into myopic demand, mispricing hedging demand, and inflation hedging components.
- Proved the Verification Theorem by applying the generalized multidimensional Itô's formula for jump-diffusions and establishing the uniform integrability of the local martingale component to rigorously validate that the candidate solution strictly coincides with the unique value function.
- Established the sufficient conditions for global uniqueness by constructing a strictly negative-definite Hessian matrix, proving the candidate strategy is the unique global maximizer of the HJB operator.

Working Experience

Vietnam Maritime Commercial Joint Stock Bank <i>Risk Modelling Associate (Dev)</i>	<i>Hanoi, Vietnam</i> <i>Jul 2025 – Present</i>
Develop quantitative credit scoring and customer rating frameworks by extracting predictive signals from large-scale behavioral, transactional, and financial customer data, supporting risk-based lending decisions and portfolio management.	
EY Consulting Vietnam <i>Credit Risk Consultant Intern</i>	<i>Hanoi, Vietnam</i> <i>Jun 2024 – Sep 2024</i>
Conducted an extensive review of academic and industry literature to develop a quantitative ESG assessment methodology, translating sustainability principles into measurable scoring criteria for corporate risk evaluation.	

Publication and Working Papers

Xuan Ngo, **Kim-Anh Le**, Thanh Nguyen, Khoa Nguyen (2025). **Does Fintech curb bank risk taking? Insights from an emerging economy**, *Finance Research Letters*, 86(F). ISSN: 1544-6123. <https://doi.org/10.1016/j.frl.2025.108722>

Van-Quy Nguyen **Kim-Anh Le**. **Wariness, Bequests, and Capital Accumulation in an Overlapping Generations Economy**. *Working Paper*.

Van-Quy Nguyen **Kim-Anh Le**. **Existence and Regularity of Competitive Equilibria in Economies with Externalities**. *Work in Progress*.

Kim-Anh Le. **Optimal Consumption and Investment Strategies with Habit Formation under Mispricing and Jump Diffusion**. *Work in progress*.

Honors and Awards

- 2-time winner, Third Prize in Calculus (Category A) at the Vietnamese Mathematical Olympiad Competition for Undergraduates (2023, 2024)**: awarded by the Vietnam Mathematical Society.
- 3-time winner, Academic Excellence University Scholarship (Fall 2023 – Spring 2024 – Fall 2024)**: top 5 of students in Mathematical Economics majors.
- President’s Certificate of Honors for achieving the national awards (2024)**: awarded by National Economics University, Vietnam.
- Certificate of Merit for Academic Research Achievements (2023)**: awarded by the Dean of the Department of Mathematical Economics.

Skills

Programming	Python, R, MATLAB
Frameworks	Pandas, NumPy, SciPy, Scikit-learn, PyTorch, Statsmodels, Matplotlib
Tools	LaTeX, SQL
English	IELTS 7.0

References

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